

Jaimin Rana

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SUMMARY

Python Developer with 3 years of experience building production-grade backend systems and AI-powered applications in the healthcare domain. Hands-on expertise in fine-tuning NLP/NER models, Retrieval-Augmented Generation (RAG), LLM integration, and clinical data pipeline automation. Contributed end-to-end to a HIPAA-relevant PHI de-identification system — from model fine-tuning (GLiNER) to production deployment — achieving 90–95% accuracy. Strong foundation in scalable API design, Redis caching, Docker, and AWS.

EXPERIENCE

Software Engineer – Backend & AI/ML

July 2023 – Present

AGS Health Pvt. Ltd., Ahmedabad

- Fine-tuned and deployed a **GLiNER NER model** for PHI de-identification in clinical documents, achieving **90–95% entity-detection accuracy** in production across patient records.
- Designed a **hybrid de-identification pipeline** combining the fine-tuned NER model with a regex-based rule engine (covering dates, addresses, hospital names, and 10+ PHI entity types), reducing false negatives and cutting processing time significantly.
- Built **Python-based automation workflows** (Flask, Pandas, NumPy) that eliminated 50% of manual data processing across clinical data pipelines.
- Built a production-grade **domain-specific search engine** over 500k+ health records, achieving sub-200ms query latency via optimized indexing and Redis caching — reducing DB load by 50%.
- Designed and maintained REST APIs (Python/Flask and Node.js/Express), improving throughput by 30% and reducing average response latency by 25% through query optimization and caching strategies.
- Orchestrated ML data pipelines using **Prefect**, enabling reliable scheduling, retry logic, and observability for automated ETL and reporting workflows.
- Containerized services with **Docker** and implemented CI/CD pipelines, enabling zero-downtime deployments and rapid iteration on AI features.
- Optimized **MongoDB** and **MySQL** schemas and queries for high-volume clinical data, reducing report generation time by 55%.

TECHNICAL SKILLS

- Languages:** Python (primary), JavaScript, Java, C++
- AI/ML/GenAI:** GLiNER, NER Fine-tuning, LangChain, OpenAI API, RAG (Retrieval-Augmented Generation), Prompt Engineering, Hugging Face Transformers, scikit-learn, spaCy
- NLP & Clinical AI:** PHI De-identification, Regex-based Rule Engines, Named Entity Recognition, Clinical NLP
- Data & Automation:** Pandas, NumPy, Prefect, ETL Pipelines, Excel Automation
- Backend:** Flask, Django, FastAPI, Node.js, Express.js, REST APIs
- Databases:** MongoDB, MySQL, Redis, Mongoose, Sequelize
- Cloud/DevOps:** Docker, AWS (S3, EC2, SQS, CodeCommit), CI/CD
- Frontend:** React.js, HTML, CSS, JavaScript
- Tools:** Git, GitHub, Linux, Postman, DSA

PROJECTS

PHI De-identification System – Clinical NLP Pipeline

2024 – Present

- Contributed end-to-end to a **HIPAA-aligned PHI de-identification system** for clinical patient documents at AGS Health, spanning data preparation, model fine-tuning, evaluation, and production deployment.
- Fine-tuned a **GLiNER (Generalist NER)** model on domain-specific annotated clinical data, achieving **90–95% entity detection accuracy** across PHI categories including patient names, dates, addresses, and provider information.
- Built a **hybrid pipeline** combining the fine-tuned NER model with a custom regex rule engine targeting structured PHI patterns (dates, phone numbers, hospital names, zip codes), improving recall and reducing end-to-end processing time by pre-filtering high-confidence deterministic entities before model inference.
- Containerized the inference service with **Docker** and deployed to production, supporting real-time de-identification of clinical documents at scale.

HealthQuery – AI-Powered Medical Search Engine

Jul 2024 – May 2025

- Built a production-grade semantic search system over **500k+ health records**, delivering **sub-200ms query responses** using optimized full-text indexing and Redis caching.
- Designed a flexible schema supporting dynamic filters, range queries, and multi-field search — architected to support future LLM-augmented query understanding.
- Reduced database load by **50%** through intelligent caching of frequent query patterns, improving system reliability under peak traffic.

- Built a **RAG pipeline** using LangChain and OpenAI GPT models to answer natural language queries over private document collections.
- Implemented chunking strategies, embedding-based retrieval, and prompt templating to improve answer accuracy and reduce hallucinations.
- Exposed the pipeline as a **FastAPI** REST service, enabling integration with front-end interfaces and downstream systems.

ownCloud – Scalable File Storage & Collaboration Platform

Jan 2025 – Present

- Architected a backend with a flattened directory model, cutting file retrieval latency by **35%** compared to nested-tree approaches.
- Integrated **Redis** for metadata and directory listing caches, significantly reducing repeated I/O overhead.
- Containerized services with Docker; deployed backend on Render and frontend on Vercel for scalable, low-maintenance production hosting.

EDUCATION

B.E. in Computer Engineering
Sardar Vallabhbhai Patel Institute of Technology, Vasad

2019 – 2023
CGPA: 8.41 / 10

H.S.C. (Science Stream)
Bits Education High School, Khambhat

2017 – 2019
83.47%

CERTIFICATIONS & AWARDS

- **RockStar Award** – Recognized for outstanding performance in Software Engineering at AGS Health Pvt. Ltd. (Q1 2024)
- **Full-Stack Web Development** – MERN stack certification by Angela Yu (2023)
- **Advanced Backend Development** – Professional Node.js course by Anurag Singh (2025)
- **LangChain & LLM Applications** – (Add any AI/ML course you’ve completed or are actively pursuing)